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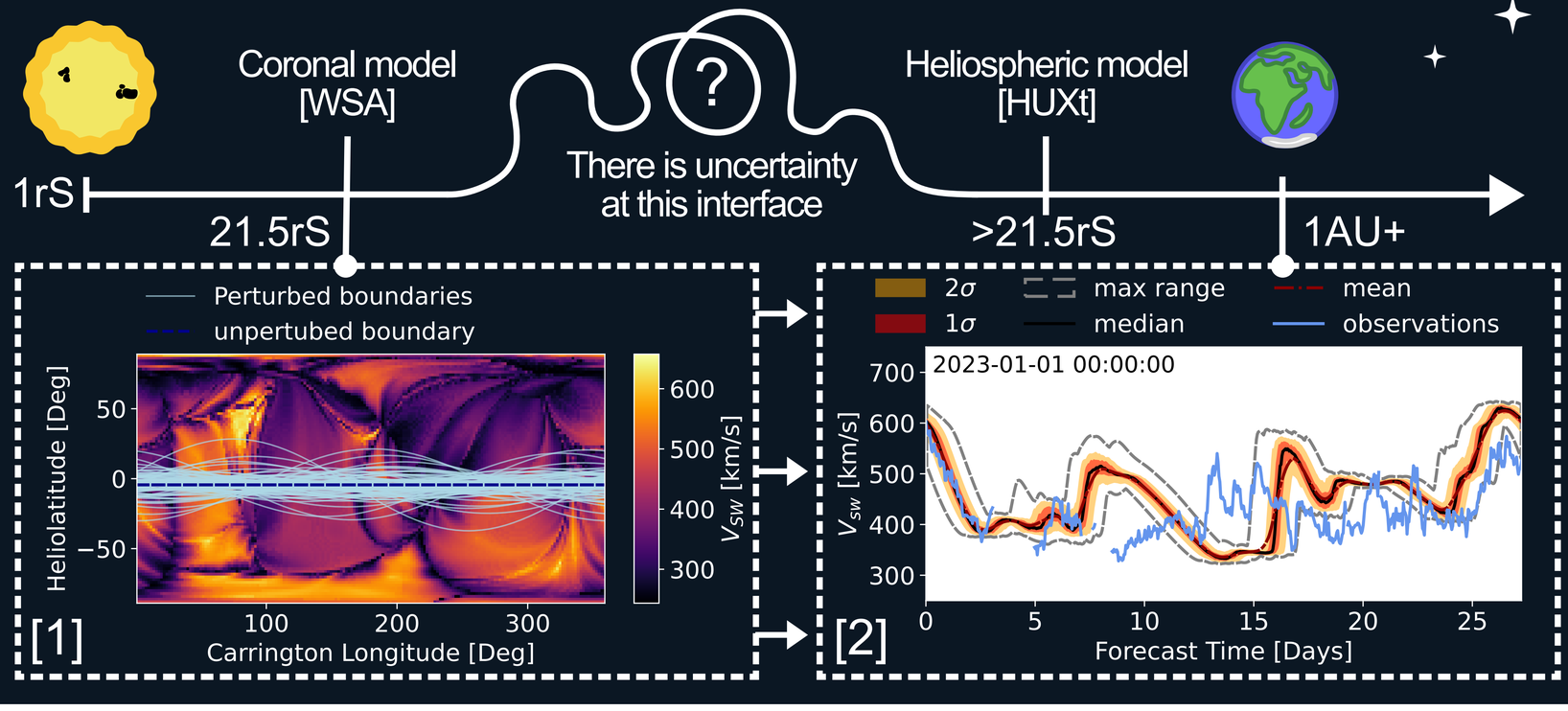
Calibrating probabilistic solar-wind forecasts driven by the Wang-Sheeley-Arge model

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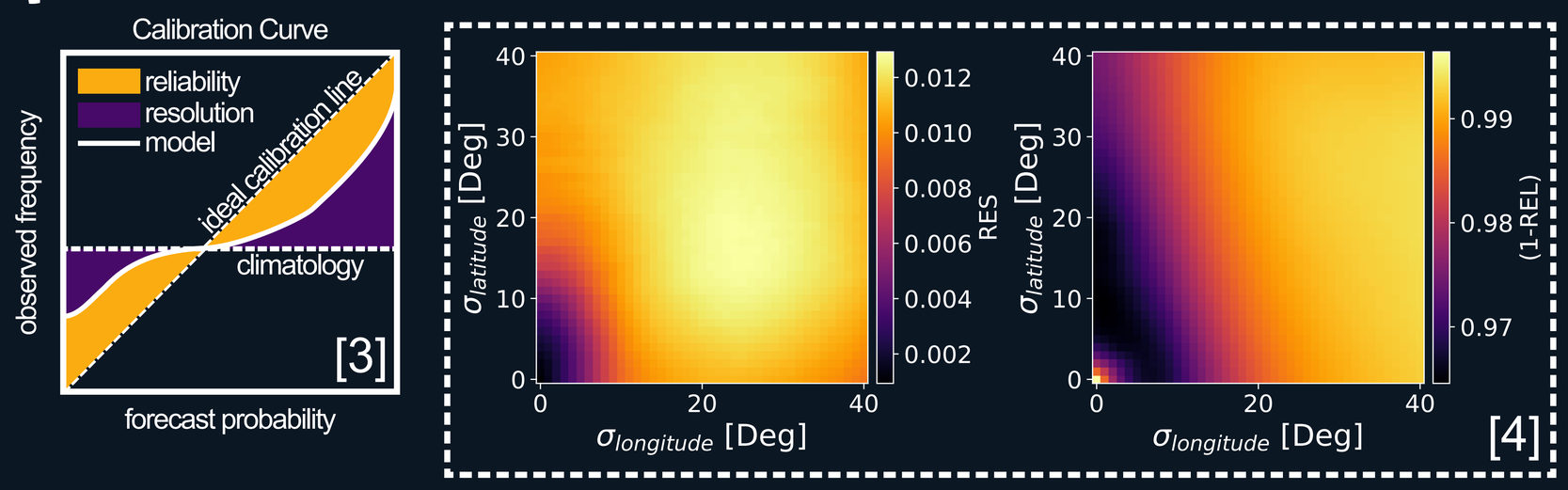
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Ensemble modelling



- **The Wang-Sheeley-Arge (WSA) model is widely used by the space weather community.**
- **We use WSA to provide inner-boundary conditions at 21.5rS** (figure [1]) to drive a heliospheric model called HUXt which models flow out to 1AU, see figure [2].
- **The solar wind estimates at 21.5rS are uncertain.**
- **We use ensemble methods to characterise the uncertainty.**
- **We can spatially perturb** (rotate) the inner-boundary derived from WSA **to sample some of this uncertainty**, see figures [1+2].

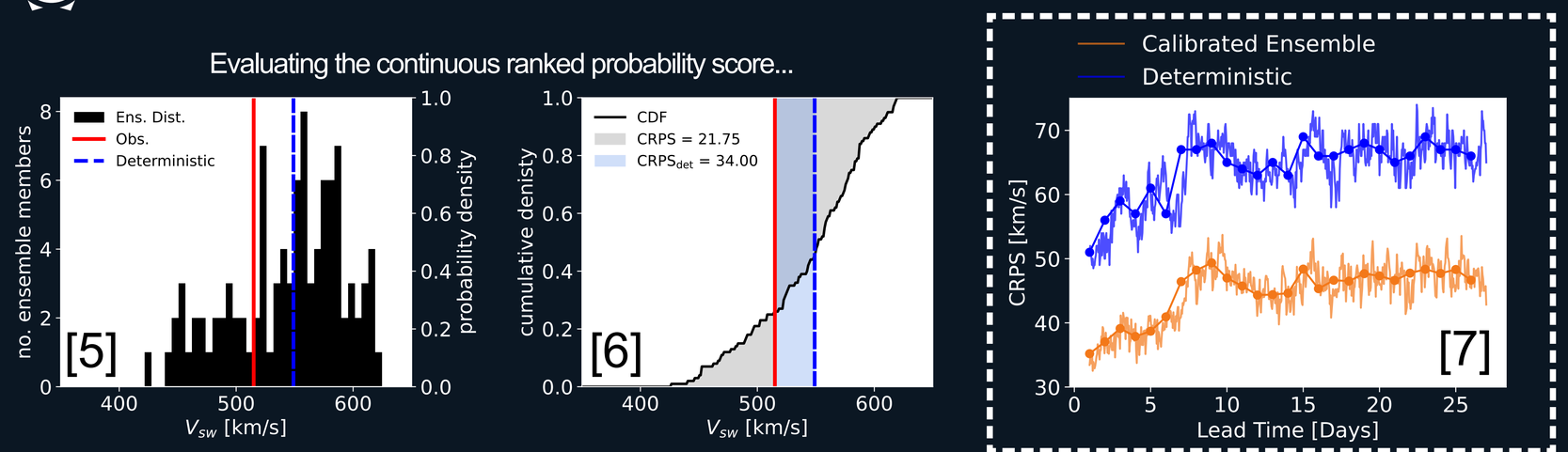
Calibration



- **A calibration curve shows how calibrated a probabilistic forecast is, see figure [3].**
- **Brier score** measures mean square error for probabilistic predictions and **can be split into reliability [REL] and resolution [RES] components** which relate to the calibration curve, see figure [3].
- Over a case study of daily WSA-HUXt ensembles from July-Dec 2020 and Jan-Dec 2023, **REL and RES often worked against each other in lower variability ensembles,** see colour maps in figure [4]**.**

- **Ensemble calibration aims to bring forecast probabilities in line with the observed frequency of events.**
- The spread of input conditions greatly impacts the ensemble calibration.
- **Increasing the spread to improve forecast reliability** (how well it reflects real-world conditions) **can reduce how clearly it distinguishes between possible outcomes,** see figure [4].
- **We fine tune the spread to strike a balance** between reliability and resolution, helping make forecasts both accurate and useful.

Forecast evaluation



- CRPS is the enclosed area between the CDF defined by the ensemble and an observation (figures [5+6]).
- **Calibrated ensemble members** carry information about likely outcomes and **add useful information content into a forecast.**
- When compared to a deterministic forecast over the case study period, calibrated ensembles reduce the average amount and variability of forecast error, see figure [7].

- We use the continuous ranked probability score (CRPS) to show that **the ensemble provides information which improves the forecast.**
- CRPS can be thought of as Mean Absolute Error as applied to a distribution of predictions.
- **A calibrated ensemble reduces the CRPS/absolute error**, see figure [7].
- A Cost/Loss analysis shows the **ensemble can provide equal/more value than a single deterministic WSA-HUXT forecast.**

References

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